

Pinsara Weerasinghe

he/him/his • Colombo, Sri Lanka (Open to Relocation) • +94719596692 • pinsarakweerasinghe@gmail.com •
linkedin.com/in/pinsara • github.com/PinsaraKWeerasinghe • pinsara.com

Senior Backend Engineer | Java & AWS Cloud Microservices

University of Moratuwa Computer Science and Engineering Honors graduate with 6 years of experience architecting high-performance backend systems across the Fintech and Aviation domains. Spent 2.5 years in Fintech delivering low-latency Java and Spring Boot backend infrastructure to optimize transactional data processing. Advanced into Aviation tech for 3.5 years, designing cloud-native microservices and enterprise data hubs on AWS utilizing the Serverless Framework and AWS CDK. Combines deep server-side logic mastery with automated cloud provisioning to deliver fault-tolerant, high-throughput microservice architectures. Seeking to leverage scalable system design and cloud engineering expertise as a Senior Backend Engineer.

WORK EXPERIENCE

Serverless Guru • Remote • 02/2023 - Present

An advanced, serverless-first AWS cloud consultancy specializing in enterprise serverless architecture, microservices, and high-scale cloud transformations for global enterprise organizations and high-growth startups.

Senior Software Engineer / Cloud Consultant • Full-time

- Architected and engineered scalable serverless microservices and a central cloud Data Hub for **Air Canada** using the Serverless Framework, Node.js, and AWS Lambda to optimize airline data workflows.
- Designed high-throughput, real-time event ingestion data pipelines integrating **Apache Kafka** topics and **AWS Glue** ETL jobs to handle complex enterprise streams seamlessly into **Amazon S3**.
- Managed distributed search indexing and analytical querying patterns by designing and executing high-performance data pipelines into **Amazon OpenSearch Service**.
- Led a major corporate infrastructure database migration from **MySQL to PostgreSQL**, maintaining strict relational transaction limits and zero-downtime structural integrity.
- Constructed programmatic system monitoring dashboards and automated anomaly alert triggers using **Amazon CloudWatch**, drastically optimizing operation costs and lowering cloud resource footprints.
- Implemented low-latency internal data access bridges by developing **GraphQL APIs** to seamlessly unify external third-party enterprise storage hubs like SharePoint.

GTN Technologies (Previously Global Market Technologies) • Colombo, Sri Lanka • 06/2019 - 02/2023

A global fintech leader and institutional trading infrastructure provider connecting banks, asset managers, and wealth platforms to over 90 global capital markets via advanced e-trading engines.

Software Engineer • Full-time

- Engineered core real-time market data extraction components and OHLC layers to parse streaming feeds from stock exchanges.
- Designed data dissemination backend engines using **Java, Spring Boot, and WebSockets/TCP Sockets** for concurrent enterprise clients.
- Optimized text-search indexing speeds across database nodes by architecting wrappers around **Apache Solr** instances.
- Built asynchronous message queues using **Apache Kafka, AWS SQS, and AWS SNS** to handle background jobs.

Intern Software Engineer • Full-time

- Engineered automated log analysis and anomaly detection utilities for real-time market price backends while executing load/stress testing to optimize data throughput and platform stability under peak traffic spikes.

EDUCATION

Bachelor of Science (B.Sc. Hons) in Computer Science and Engineering

University of Moratuwa • Moratuwa, Sri Lanka • 09/2016 - 03/2021

* Completed intensive 4-year professional engineering curriculum specializing in advanced software architecture, data structures, cloud systems, and distributed computing databases.

G.C.E. Advanced Level Examination in (Physical Science Stream (Combined Mathematics, Physics, Chemistry))

Mo/Malwaththawala National School • Wellawaya, Sri Lanka • GPA: AAA • 01/2001 - 08/2014

* Achieved straight 'A' passes across all core technical subjects. Secured a highly competitive Z-Score of 2.3135, earning a District Rank of 2 out of thousands of candidates nationwide.

PROJECTS

Meridiano Ceylon – Tourism Booking Platform (www.meridianoceylon.com) • 04/2026 - 07/2026

Independent / Commercial Project

- **Technologies Used:** Next.js, React, TypeScript, Firebase, Tailwind CSS
- Architected a production Next.js travel platform with integrated Firebase cloud services—delivering dynamic tour package management, real-time booking enquiries, user authentication, and secure media asset storage.

Data Analytics & ML Platform • 05/2020 - 03/2021

University of Moratuwa (Final Year Research Project)

- **Technologies Used:** Apache Spark, NLP, Python, Machine Learning, Data Analytics
- Developed a final-year ML analytics framework using NLP and Apache Spark to process large-scale driver telemetry and feedback, computing real-time QoS scores and dynamic rankings.

MAS Holdings Quality Process Automation • 10/2020 - 12/2020

Fcode Labs

- **Technologies Used:** Flutter, Dart, Firebase
- Engineered a cross-platform Flutter/Dart app with Firebase integration to digitize garment evaluation workflows—automating scheduling, measurement logging, and photo capture—while developing a smart fashion tool to capture and visually compare clothing items.

VOLUNTEERING & LEADERSHIP

Department of Computer Science & Engineering, University of Moratuwa

Organizing Committee Member

Coordinated event logistics, technical showcases, and visitor engagement for Moratuwa's flagship CSE exhibition.

SKILLS

Cloud & Serverless Architecture: AWS (Lambda, CloudWatch, DynamoDB, Event-Driven Architecture, Glue, Microservices Architecture, OpenSearch, REST & GraphQL APIs, S3, Serverless Framework, SNS), SQS

Programming Languages: Dart, Java, JavaScript, Node.js, Python, SQL, TypeScript

Frameworks & Search Engines: Apache Solr, Express.js, Nest.js, Next.js, React, Spring Boot

Databases & Data Streaming: Amazon DynamoDB, Apache Kafka, Firebase, MySQL, PostgreSQL

DevOps, CI/CD & Tools: AWS CloudWatch Dashboards, Bamboo, Bitbucket, Docker, Git, Litestream

Personal Interests: Automotive Engineering & Mechanical Restorations, Open-Source Cloud Tooling, Travel & Outdoor Exploration